

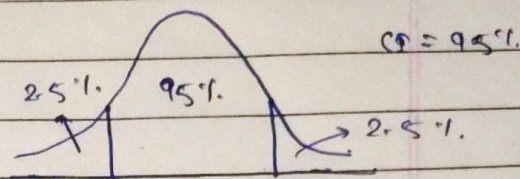
→ Confidence Interval and Margin of Error

Point Estimate

$$[\bar{x}] \rightarrow [\mu]$$

$$\bar{x} = 2.5$$

$$\mu = 3$$



→ Confidence Interval:

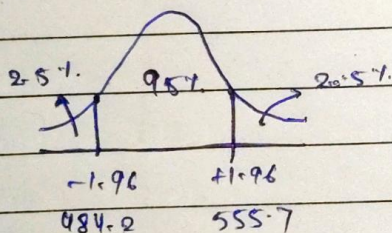
Point Estimate $\pm$ Margin of Error

$$Z\text{-test} \Rightarrow \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \left\{ \begin{array}{l} \text{If } t \text{ test then use} \\ t \text{ in place of } Z \end{array} \right.$$

Eg: On the verbal section of CAT exam, the standard deviation is known to be 100. A sample of 30 test takers has a mean of 520. Construct 95% CI above the mean.

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\alpha = 0.05$$



$$1 - 0.025 = 0.9750$$

$$\pm 1.96 \Rightarrow Z\text{-table}$$

$$\text{Lower CI} = 520 - (1.96) \frac{100}{\sqrt{30}} = 484.2$$

$$\text{Higher CI} = 520 + (1.96) \frac{100}{\sqrt{30}} = 555.7$$

Conclusion:

We are 95% confident about the mean CAT score is between 484.2 and 555.7